

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEE 325 — Past Questions & Materials

by Renance

Course: PEE 325

Institution: Federal University of Petroleum Resources, Effurun

Papers in this pack: 1

Levels: 300

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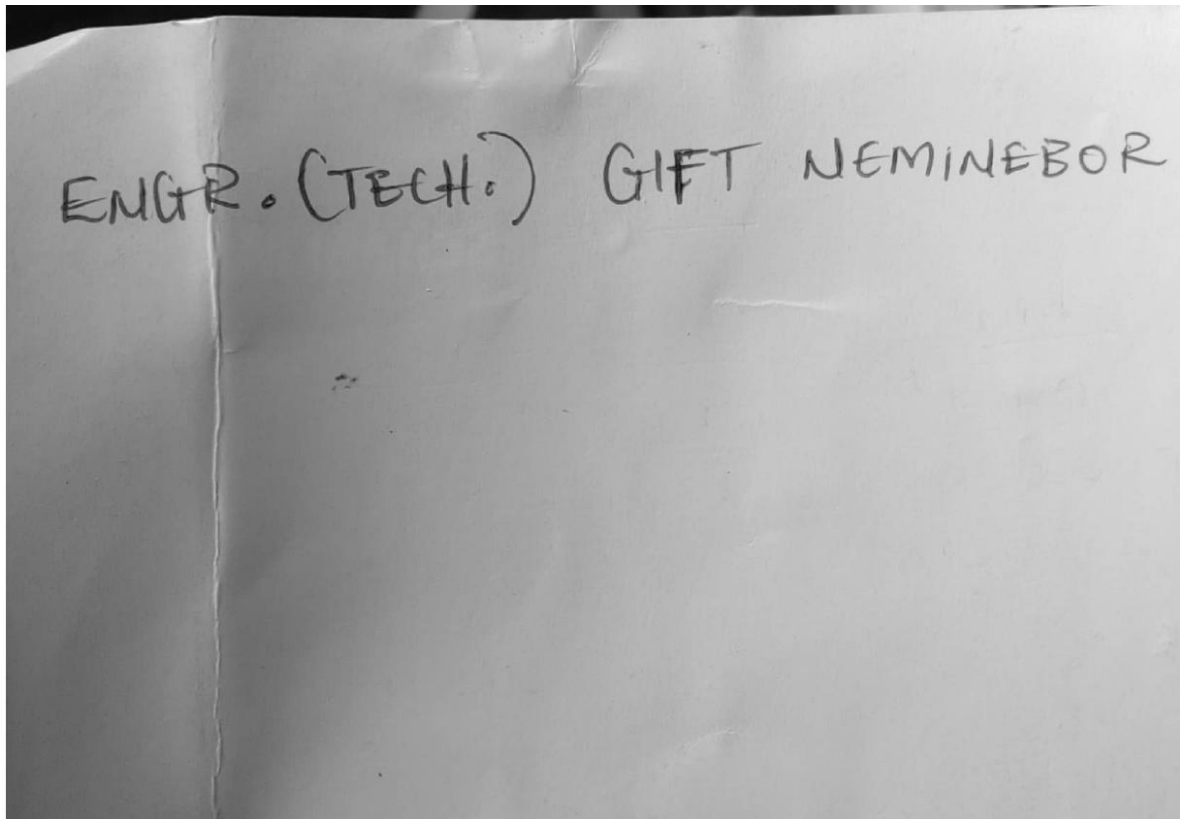
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Paper 1 of 1 — PEE 325

Level 300 • Second Semester • Handout

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PEE 325 (PRACTICAL)

Importance DRILLING FLUID ANALYSIS

- Removing Cuttings
- Stopping of water into the hole
- * Collapse of hole

Functions of Drilling fluids

- ① Removal of Cuttings
- ② Control formation pressure
- ③ Maintain wellbore stability
- ④ Seal permeable formations
- ⑤ Cool and lubricate the bit
- ⑥ Transmit hydraulic energy to lift and down hole tools
- ⑦ Aides in well data analysis
- ⑧ Minimize reservoir damage

* Mud logger + Examine Cuttings (rocks) from the reservoir

Base - fluid

Types of Drilling fluids

- ① Water base $\begin{cases} \text{fresh water} \\ \text{sea water} \end{cases}$ Water based mud is always problematic
- ② Oil base — Diesel
- ③ Synthetic — fluids

* Water based mud cannot withstand high pressure and high temperature.

But oil based mud can withstand any pressure or temp.

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* Do not use water based mud to drill through Shale formations.

Properties of Drilling Mud

① Mud density

$H_2O + \text{Clay (Bentonite)} = \text{Drilling mud}$
 + (Barite (to increase its weight), $CaCO_3$, Haematite)

Weighting materials

② Viscosity

Less Viscous fluids flow faster

Why water flows Because of change in elevation.

TO increase the Viscosity

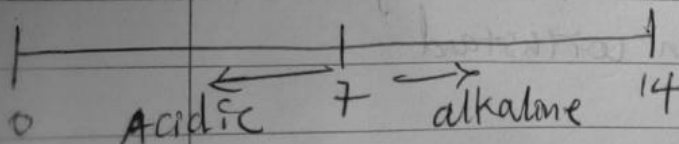
- Bentonite
- Viscosifiers e.g.

And the examples

What are additives — additives are other things to add apart from the major components.

③ PH Control additives

(PH value of drilling mud affects Corrosion of drill strings and tools)



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④ Filtration (fluid Loss)

⑤ Sand Control

PH Value for Standard drilling Mud (API specification)
is $(8.5 - 12.5)$.

PEE 325 (per Lab)

28th/01/2022

1) State 10 functions of Drilling fluid → Quiz

The most important function of drilling fluid to mud loggers is

⇒ Assist in the collection and interpretation of information available from drill cuttings, cores and electrical logs.

Calcrete	0	
Dolomite		
Limestone	2,000	0.49 psi/ft
Siltstone	3,000	0.50 psi/ft
Coarse Sandstone	7,000	0.51 psi/ft
Shale	8,000	
Sandstone	10,000	

If mud weight = 8.000 pounds per gallon

Assignment (find the meaning) of the following $P_h = 0.052 \times MW \times TVD$

① Wildcat well

② Exploratory well

③ Appraisal well

$$P_h = 4160 \text{ psi}$$

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EXPERIMENT 1MUD WEIGHT DETERMINATION

Importance $\frac{1}{2}$ (To control formation pressure)

$$\text{To Calculate mud gradient} = \frac{P_h - \text{hydrostatic pressure}}{\text{TVD}} = \frac{4160 \text{ psi}}{10,000 \text{ ft}} = 0.416 \approx 0.42 \text{ psi/ft.}$$

$P_h \rightarrow$ Hydrostatic pressure

$P_f \rightarrow$ formation pressure

At all times $P_h \geq P_f$

Kick - Influx of formation fluid into the wellbore

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}}$$

Well Control (Types)

- ① Primary — Is the use of drilling mud to control formation pressure
- ② Secondary — use of Blowout preventers
- ③ Tertiary — Every other parameter or technique used apart from primary and Secondary

* Degasser (Poor boy)

* When the weight/density of the mud decreases, you add weighting materials +

Examples — CaCO_3 , Barite

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Group 8 Elements

1000ml $\rightarrow$ 1ltr

Rare - Because they are scarce

Inert - They do not react with other elements

Noble - Because they are rare

Materials Needed +

① Water (freshwater)

② Clay (Bentonite)

③ Weighing material (Bentonite)

Measuring Equipment

① Mud Balance

② Measuring Cylinder

Equipment required +

① Mud Mixer

② Weighing Balance

③ Petri dish

Bentonite + water = will

rise, swell and give the water

viscosity but does not add

weight to it.

5 x 17.5 g of Bentonite Clay

Empty Petri dish = 45.2g

5 x 350ml of freshwater

87.5

17.5 = 22.5g

132.7g

87.5g + 1750mls.

The clay with

white colour - Bentonite

Brown colour - Bentonite.

* Machine balances/absorbs

the weight of the petri-dish

* The machine stirs for 5 mins.

Precautions

① The table must be horizontally balanced

② The Cup must be closed

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③ The Cup must be filled to the brim

Specific gravity $\rightarrow 1.0$ 87.5
87.5

Pounds per gallon $\rightarrow 8.3 \text{ ppg}$ 175.0

For Standard water +
 $\rightarrow 8.33 \text{ ppg}$ - density of water
 $\rightarrow 1.00$ - specific gravity

Precaution It was ensured the weighting balance was clean in order to have a precise/accurate result.

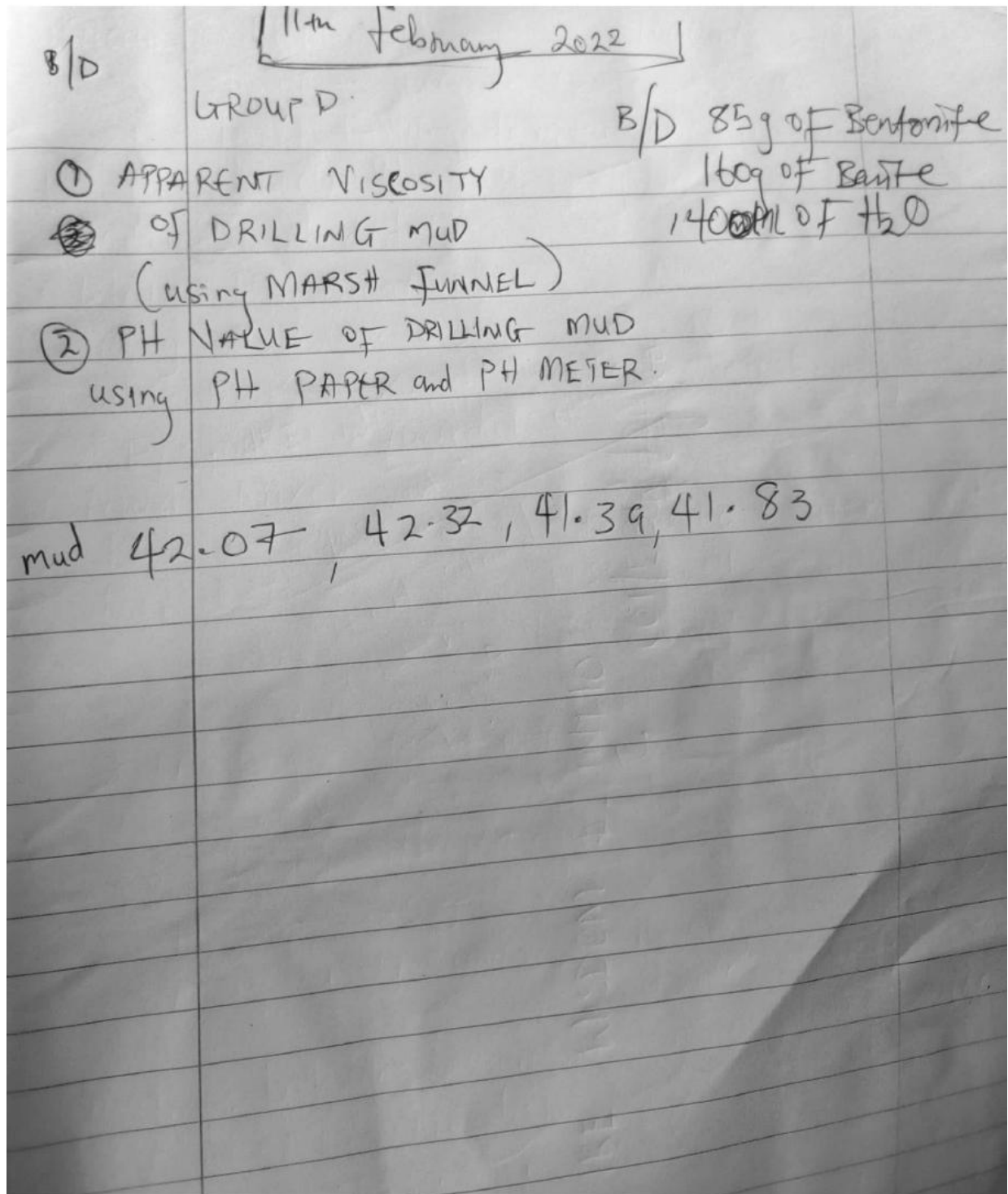
Sample

① fresh water 8.33 ppg
 ② H_2O + Bentonite only 8.50 ppg
 ③ H_2O + Bentonite + Barite 9.15 ppg

* As mud weight increases, the pump force needed, will also increase

PPG	Lb/ft ³	psi/1000ft
8.33	62.5	435
8.50	63.8	443
9.15	68.6	478

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